

Sample Questions

1. Objective questions:

True/False Questions

T F All objects in Java are allocated from the heap and accessed through reference variables

Matching terms with descriptions:

Lisp, Functional languages...

_____Artificial intelligence

_____Programming consists of building the function that computes the answer.

2. Short essay questions

- Explain the two problems with abstract data types that are ameliorated by inheritance.
- Discuss why OO languages support dynamic binding.

3. Code-related questions

1) Answer the questions with regards to the following source code.

```
class Animal {
    Animal(int i) {
        System.out.println("Animal constructor");
    }
}

class Cat extends Animal {
    Cat(int i) {
        super(i);
        System.out.println("Cat constructor");
    }
}

public class Tiger extends Cat {
    Tiger() {
        super(9);
        System.out.println("Tiger constructor");
    }
    public static void main(String[] args) {
        Tiger x = new Tiger();
    }
}
```

What is output of the program? Explain how you reach your answer.

4. Code-related question

A Maryland license plate consists of 7 upper case letters or numbers, for example, 1AB2345 .

Write a BNF grammar that describes MD license plates.

5. Code-related question

Consider the following grammar:

$$\langle S \rangle \rightarrow a \langle S \rangle c \langle B \rangle \mid \langle A \rangle \mid b$$
$$\langle A \rangle \rightarrow c \langle A \rangle \mid c$$
$$\langle B \rangle \rightarrow d \mid \langle A \rangle$$

Which of the following sentences are in the language generated by this grammar? Show a leftmost derivation for the sentence in the language.

a) abcd

b) acccb